

NAMIBIA SENIOR SECONDARY CERTIFICATE

CHEMISTRY ORDINARY LEVEL

6117/1

PAPER 1 Multiple Choice

45 minutes

Marks 40

2023

Additional Materials: Multiple choice answer sheet
Non-programmable calculator
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS AND INFORMATION TO CANDIDATES

- Write in soft pencil.
- Make sure that you receive the multiple choice answer sheet with **your examination number** on it.
- There are **forty** questions on this paper. Answer **all** questions.
- For each question, there are four possible answers **A, B, C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the separate answer sheet.
- If you want to change an answer, thoroughly erase the one you wish to delete.
- The Periodic Table is printed on page 11.
- **Read the instructions on the answer sheet carefully.**
- Each correct answer will score one mark.
- Any rough working should be done in this booklet.
- You may use a non-programmable calculator.

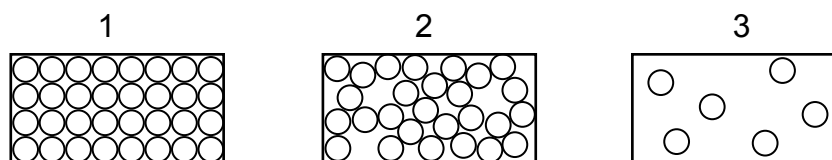
This document consists of **11** printed pages and **1** blank page.



Republic of Namibia

MINISTRY OF EDUCATION, ARTS AND CULTURE

- 1 Diagrams 1, 2 and 3 represent particle arrangements in three states of matter.

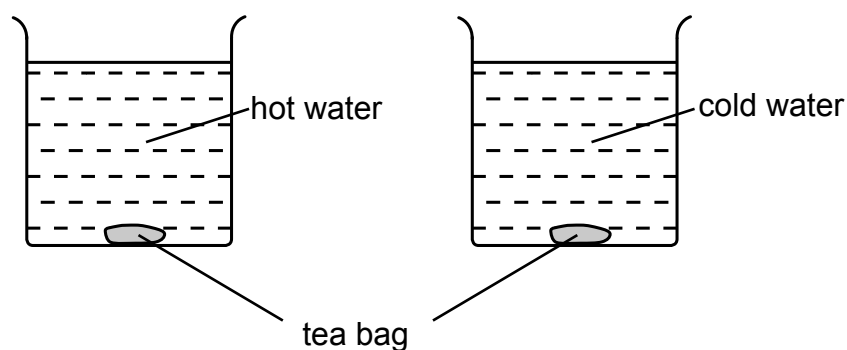


Which statement correctly compares the kinetic energy of particles?

- A 1 has higher kinetic energy than 2
 B 2 has lower kinetic energy than both 1 and 3
 C 2 has higher kinetic energy than 3
 D 3 has higher kinetic energy than both 1 and 2
- 2 A certain food colouring contains a dissolved mixture of red and yellow dyes. It is suspected that the mixture was contaminated with a third dye X.

Which method could be used to investigate the presence of the third dye?

- A chromatography
 B distillation
 C evaporation
 D filtration
- 3 A tea bag was added to each of the beakers shown in the diagram. One beaker contained hot water and the other beaker contained cold water. In both beakers the colour spreads out.



Which result and explanation are correct?

	result	explanation
A	colour spreads faster in cold water	particles move faster at a higher temperature
B	colour spreads slower in cold water	particles move slower at a higher temperature
C	colour spreads faster in hot water	particles move faster at a higher temperature
D	colour spreads slower in hot water	particles move slower at a higher temperature

4 Which expression gives the number of nucleons in A_ZX ?

- A A
- B Z
- C $A - Z$
- D $Z - A$

5 An element Y is in Group V of the Periodic Table.

Which of the following is true about the atomic structure of element Y?

- A It has 5 occupied shells.
- B It has 5 electrons in the outer shell.
- C It has 5 protons in its nucleus.
- D It has 5 neutrons in its nucleus.

6 Which metal forms a hydroxide which is very soluble in water?

- A calcium
- B copper
- C lithium
- D magnesium

7 Bromine is found in group VII of the Periodic Table.

Which row is true about the colour and the physical state of bromine at room temperature?

	colour	physical state at room temperature
A	red-brown	solid
B	red-brown	liquid
C	yellow-green	gas
D	yellow-green	liquid

8 Which name is given to mixtures of a metal with other metals?

- A alloy
- B compound
- C element
- D ore

9 Which substance has the correct type of bonding?

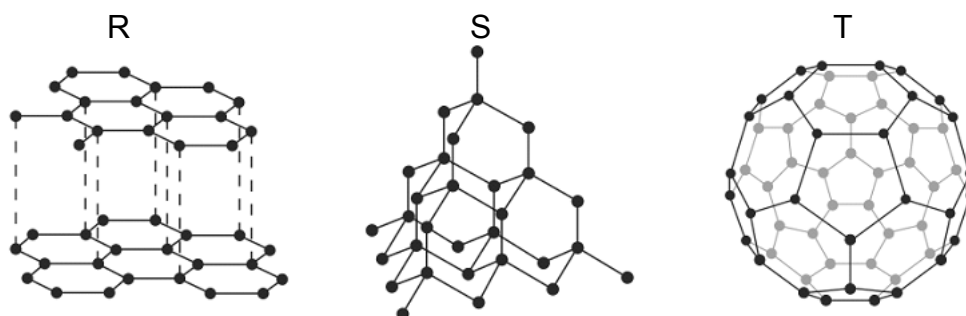
	substance	type of bonding		
		ionic	covalent	metallic
A	carbon dioxide	✓	×	×
B	magnesium bromide	×	×	✓
C	sodium	✓	×	×
D	oxygen	×	✓	×

10 Lithium atoms react with chlorine atoms to form lithium chloride.

Which statement is correct?

- A Lithium loses one electron to form a negative charge.
- B Lithium gains one electron to form a positive charge.
- C Chlorine loses one electron to form a negative charge.
- D Chlorine gains one electron to form a negative charge.

11 R, S and T represent three different structures of an element.



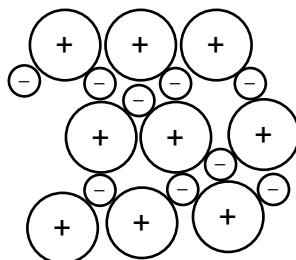
Which structures are giant covalent lattices?

- A R, S and T
- B R and S only
- C R and T only
- D S and T only

12 Which of the following explains why metals have a high melting point?

- A Their atoms are regularly arranged in layers.
- B They have a lattice of positive ions.
- C They have free moving electrons that can carry charge.
- D They have strong electrostatic forces.

13 The diagram represents the generalised structure of a substance.



What is the substance?

- A chlorine
- B sodium
- C sulfur dioxide
- D xenon

- 14 The word equation shows the reaction between hydrogen and oxygen to produce water.



What is the balanced chemical equation for the reaction?

- A $2\text{H} + 2\text{O} \rightarrow 2\text{H}_2\text{O}$
 B $\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$
 C $\text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{O}$
 D $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$
- 15 Polythene is an example of a material found in the environment.

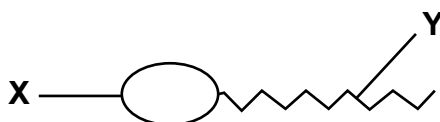
Which row is correct about the nature of polythene and the type of bonds it consists of?

	nature of material	type of bonds
A	natural	covalent
B	natural	ionic
C	synthetic	covalent
D	synthetic	ionic

- 16 Which other substance, other than clay is used to make cement?

- A limestone
 B sand
 C stones
 D water

- 17 The diagram shows the generalised structure of a soap molecule.



Which row is correct about how soap removes oily stains during washing?

	part X	part Y
A	mixes with water	mixes with water
B	mixes with water	mixes with the oily stains
C	mixes with the oily stains	mixes with the oily stains
D	mixes with the oily stains	mixes with water

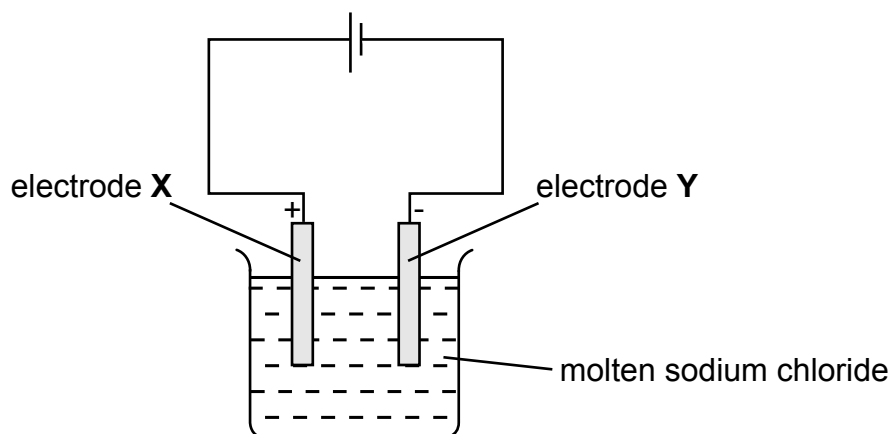
- 18 Which oxide is acidic?

- A aluminium oxide
 B magnesium oxide
 C carbon dioxide
 D potassium oxide

- 19** Naphthalene is a hydrocarbon often used in moth balls.
The percentage composition is 93.75% carbon and 6.25% hydrogen.

What is the empirical formula of naphthalene?

- A** CH
B C₃H₂
C C₄H₅
D C₅H₄
- 20** A solution of potassium carbonate, K₂CO₃, has a concentration of 0.03 mol/dm³.
Which mass of potassium carbonate is dissolved in 1 dm³ of this solution?
- A** 1.38 g
B 4.14 g
C 13.80 g
D 41.40 g
- 21** The diagram shows the electrolysis of molten sodium chloride, NaCl.



Which is correct?

- A** oxidation occurs at electrode X and the equation is: $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$
B oxidation occurs at electrode Y and the equation is: $\text{Na}^+ + \text{e}^- \rightarrow \text{Na}$
C reduction occurs at electrode X and the equation is: $\text{Na}^+ + \text{e}^- \rightarrow \text{Na}$
D reduction occurs at electrode Y and the equation is: $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$

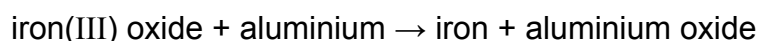
22 Which row describes the properties of a transition element?

	density	melting point	colour of compounds
A	high	low	no colour
B	high	high	blue
C	low	low	brown
D	low	high	blue

23 Which statement is true about catalysts?

- A A catalyst changes chemically after participating in a chemical reaction.
- B A catalyst increases the rate of reaction by reducing the activation energy of a reaction.
- C An inhibitor is a substance that increases the rate of a reaction.
- D Enzymes are proteins that catalyse inorganic reactions.

24 The word equation shows the reaction between iron(III) oxide and aluminium.



Which substance is the reducing agent in the reaction?

- A aluminium
- B aluminium oxide
- C iron(III) oxide
- D iron

25 Which statement(s) describes the properties of acids?

- 1 react with ammonium sulfate to form ammonia
- 2 react with metals to form salt and water
- 3 react with carbonates to form salts, water and carbon dioxide

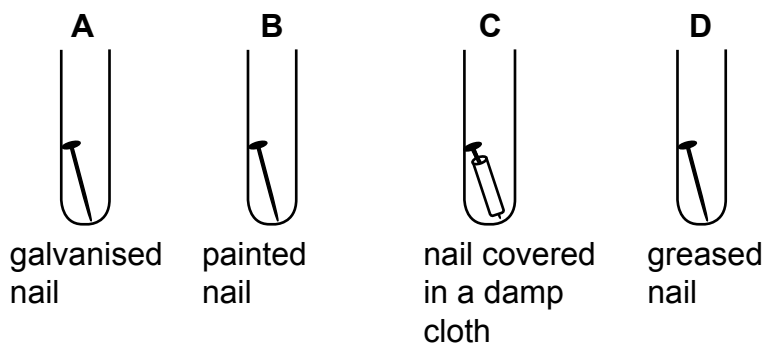
- A 1 and 2 only
- B 2 only
- C 2 and 3 only
- D 3 only

26 Which two materials are **most** suitable to prepare sodium nitrate?

- A sodium and nitrogen
- B sodium and dilute nitric acid
- C sodium hydroxide and dilute nitric acid
- D sodium oxide and ethanoic acid

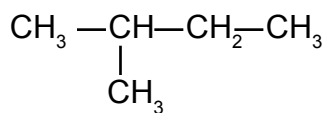
- 27 Which aqueous ion forms a white precipitate when acidified aqueous barium nitrate is added to it?
- A chloride
B iodide
C nitrate
D sulfate
- 28 Which metal is extracted from the ore, chalcopyrite?
- A aluminium
B copper
C iron
D lead
- 29 In an experiment four identical iron nails are placed in test-tubes with different conditions as shown.

Which of the following iron nails rust?

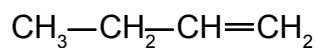


- 30 What is the general formula of alkanes?
- A $C_n H_{2n}$
B $C_n H_{2n+1}$
C $C_n H_{2n+2}$
D $C_n H_{2n+1} OH$
- 31 Which pair of compounds belong to the same homologous series?
- A CH_3CH_3 and $CH_3CH_2CH_3$
B CH_3CH_2OH and $CH_3OCH_2CH_3$
C $CH_2CHCH_2CH_3$ and $CH_3CH_2CH_2CH_3$
D CH_3CH_2OH and CH_2CHCH_2OH

- 32 The structures of two compounds P and Q are shown.



P

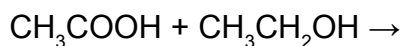


Q

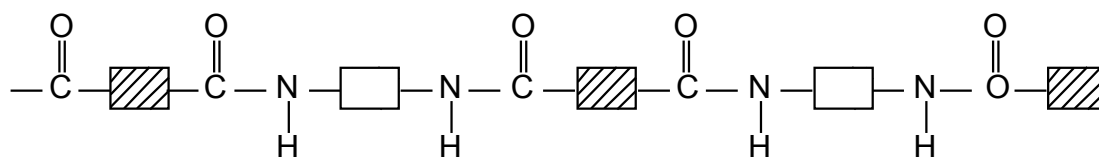
Which row about the structures is correct?

	reacts readily with bromine	forms a polymer
A	P	P
B	P	Q
C	Q	P
D	Q	Q

- 33 What is the name of the product formed when the reaction shown takes place?



- A ethyl ethanoate
 B ethyl methanoate
 C methyl ethanoate
 D methyl propanoate
- 34 The diagram shows part of a polymer



Which polymer is represented?

- A a carbohydrate
 B a protein
 C nylon
 D terylene
- 35 What is the main reason why plastic causes environmental pollution?
- A They cause acid rain.
 B They cannot be decomposed by bacteria.
 C They cause eutrophication.
 D They contribute to climate change.
- 36 Which compound causes permanent hardness in water?
- A calcium carbonate
 B calcium hydrogen carbonate
 C magnesium hydrogen carbonate
 D magnesium sulfate

37 What is the name of the method used for the commercial preparation of oxygen?

- A** diffusion
- B** distillation
- C** evaporation
- D** fractional distillation

38 Vegetation is sometimes used to form compost. This compost can be used to fertilise soils.

Which gas is likely to be present in a higher percentage during the decomposition of vegetation?

- A** carbon monoxide
- B** methane
- C** oxygen
- D** sulfur dioxide

39 When calcium carbonate is heated strongly, two substances **Y** and **Z** are formed.

Substance **Y** is a white solid that reacts with water, giving out heat.

Substance **Z** is a colourless gas.

What are substances **Y** and **Z**?

	Y	Z
A	calcium chloride	carbon dioxide
B	calcium oxide	oxygen
C	calcium oxide	carbon dioxide
D	calcium hydroxide	oxygen

40 Which of the following is a use of sulfur dioxide?

- A** manufacture of cement
- B** manufacture of detergents
- C** manufacture of food preservatives
- D** treating acidic soil

The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).

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